

Proactive Detection of Wastewater Overflows for Smart Sanitary Sewer Systems: Case Study in South Australia

Nhu Cuong Do¹; Luke Dix²; Martin Francis Lambert, M.ASCE³; and Mark Leslie Stephens⁴

Abstract: Sewage systems are built to carry contaminated wastewater from domestic discharge points to collection points for treatment. However, their capacity can be reduced by obstructions, displaced pipe joints, or broken pipes, creating abnormal hydraulic conditions. Wastewater overflows are a potential consequence of these abnormal conditions, which pose a direct threat to the environment and human health. This paper describes a permanent continuous monitoring system of a real sewage network using ultrasonic water level sensors for the purpose of blockage/choke detection, as installed in the suburb of Stonyfell, South Australia. From 62 available data sets collected over 1 year, two distinctive features of growing chokes were identified, including irregular peaks and durations that the water level remains irregularly high in the sewer maintenance holes. An early choke detection method was formulated based on the later feature, which continuously scans the near-real-time data to find time periods containing these abnormal water levels. Application of the methodology showed that the proposed method is effective in detecting possible chokes and overflow events before they occur. In most cases, the warning from the first detection is early enough to allow proactive maintenance attendance to be scheduled by the water utility. DOI: [10.1061/JWRMD5.WRENG-5589](https://doi.org/10.1061/JWRMD5.WRENG-5589). © 2022 American Society of Civil Engineers.

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Introduction

Hidden below the ground surface and often unnoticed, sewage systems play a vital role in maintaining sanitation within populated urban areas by transporting wastewater, predominantly from domestic discharges, to collection points for treatment in a closed system. This indispensable urban infrastructure provides a means of preventing environmental pollution and protection of human health from a range of potential waterborne diseases. Under normal conditions, a typical large sewage system can transfer millions of liters of wastewater on a daily basis. In the case of South Australia, for instance, the sewage system in the Greater Adelaide Region receives approximately 260 million L of wastewater every day (Department of Planning and Local Government 2010). The hydraulic capacity of the system, however, can be significantly reduced by the presence of obstructions (e.g., tree roots, the build-up of solidified fat and oil, and the intrusion of plastic bags, rags, baby wipes, and construction materials), displaced pipe joints, or broken pipes. In the case where hydraulic capacity is reduced below the discharge from the upstream catchment, wastewater can escape from the system resulting in an overflow to the

environment, or internally within customers' properties, presenting a potential threat to properties, the environment, and human health (e.g., Tchobanoglous et al. 2003; Phillips et al. 2012).

In terms of asset management, maintenance of wastewater systems to reduce and minimize wastewater overflows is an essential task. Conventionally, maintenance activities are classified into two categories: reactive and preventive maintenance. Reactive maintenance involves the use of equipment until it fails and requires replacement or repair. This includes the emergency maintenance of failed assets, which often relies on notification by customers or the community of the failure (e.g., in the case of a wastewater system either slow drainage or visible overflow). Preventive maintenance, on the other hand, comprises scheduled activities, including scheduled cleaning and inspections. This increases the reliability of the overall system.

The most common method for inspecting internal conditions (including the identification of chokes) within a sewer pipe, for preventive maintenance, is through the use of closed-circuit television (CCTV). This method uses a camera inserted into the sewer pipe (typically using either a remotely controlled robot or semirigid cable) to capture videos or a series of images. These videos and images are then assessed, either manually or through the use of analytics, to identify choke locations and assess the internal condition of the pipes. Fig. 1 shows pipes that are completely blocked by tree roots [Fig. 1(a)] and partially blocked by mud and roots [Fig. 1(b)].

The CCTV footage enables the direct observation of not only blockages but also internal conditions of the sewer pipes, allowing operators to assess the cause of a service problem and identify the appropriate remedial action required (i.e., pipe cleaning only or the repair/replacement of pipe sections as required). However, the use of CCTV cameras for sewer pipe inspection has been criticized in recent years due to a number of practical limitations. These include costly and time-consuming procedures, the need for manual image assessment and fault classification by qualified technicians, image quality dependency on camera and lighting condition, and difficulties in operating effectively under variable to high water

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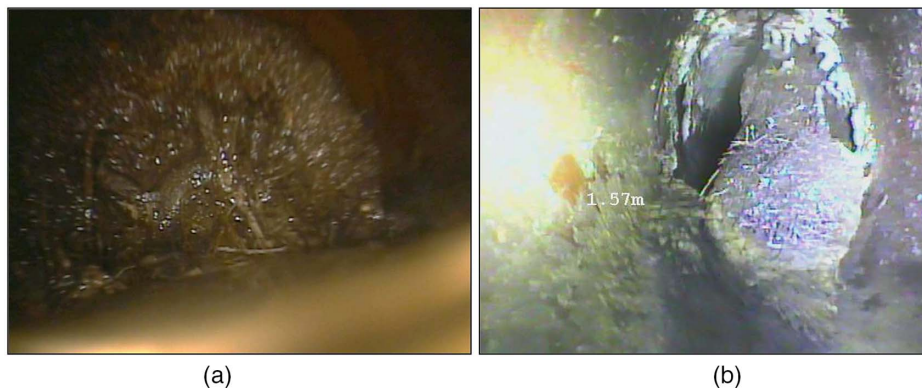


Fig. 1. CCTV footage of blockages found in SAW network pipes: (a) complete blockage caused by tree roots; and (b) partially blockage caused by mud and roots. (Images courtesy of SA Water.)

level conditions (Tscheikner-Gratl et al. 2019; Dirksen et al. 2013; Romanova et al. 2013). In addition, it has been noted that the level of misreporting and incorrect classification of pipe defects could be 20%–30% due to subjective image processing combined with human factors (Dirksen et al. 2009). That said, there have been recent advances in the automation of image assessment and classification systems using analytic methods including machine learning (e.g., Kurma et al. 2018; Stephens et al. 2020b).

The use of acoustic devices in sewer inspection and monitoring is an alternative to using CCTV cameras. For example, Bin Ali et al. (2011) proposed a technique involving the analysis of reflected acoustic signals to identify pipe structural defects and blockages created by sediment and other materials, such as rubble and fats. Romanova et al. (2013) used the analysis of acoustic intensity data recorded by a microphone array within two adjacent maintenance holes to assess the condition of the pipe between them. Although these methods offer advantages over the use of CCTV cameras in terms of earlier detection and measurement repeatability (Romanova et al. 2013), the practicality of their application in permanent monitoring and proactively identifying potential choke events remains uncertain.

Sewer failure or blockage prediction is another method of preventive maintenance, in which prediction models are often implemented on an entire sewer network to identify high-risk regions/factors relevant to asset failure. Fundamentals for these models are applications of statistical or machine learning techniques to rich historical data sets. For example, Savic et al. (2006) applied evolutionary polynomial regression to a real database containing collapse and blockage incident records for identifying the likelihood of failure types. Bailey et al. (2016) developed a predictive blockage likelihood model by applying a decision tree technique to the asset and historical incident data. Similarly, Cameron et al. (2017) applied a machine learning–based technique to the pipe characteristics, surrounding environment, and customer behavior databases for developing another predictive blockage likelihood model. Such important approaches provide insights into historical data sets and can associate different types of factors together. However, these methods often provide a coarse level of proactive maintenance and are likely suitable for relatively long-term management. Advanced techniques and technologies that can detect early signs of blockages/wastewater overflows within individual pipes or manholes to provide a more detailed level of proactive maintenance are still a practical challenge for water utilities.

The South Australian Water Corporation (SA Water) has been an early adopter of Internet of Things (IoT) technology in water and

wastewater systems (e.g., Gong et al. 2020; Stephens et al. 2020a). For wastewater in particular, SA Water has developed a smart sensing platform, from IoT sensors through to bespoke analytics, for monitoring sewer chokes in a real network through the deployment of 82 ultrasonic water-level sensors. In contrast to CCTV camera and acoustic techniques, the permanent deployment of level sensors focuses on detecting early signs of choke events through the time-series water-level data collected across the sewer system. This smart platform, as part of SA Water system, allows proactive rectification and prevention of wastewater overflows.

Although time-series data analysis for abrupt change detection is an important component in various research fields such as climatology (e.g., Reeves et al. 2007; Wang 2008), computer science and computational linguistics (e.g., Radke et al. 2005; Rybach et al. 2009), or health and medicine (e.g., Staudacher et al. 2005; Yang et al. 2006), its application in sewage-related research for blockage/choke detection at a network scale has not been found elsewhere in the literature. This is possibly because the initial cost for mass deployment of IoT sensors is significantly higher than for other reactive detection methods. In addition, data recorded through these activities is often held within the water utilities, which sometimes limits access to the data sources.

As one of the first studies exploring this research area, this paper will describe (1) the challenges of monitoring wastewater levels in sewer pipes using ultrasonic sensors that are permanently located within sewer inspection points and maintenance holes, and (2) the investigation of methods for the early detection of choke events using data from the sensing network. In particular, data associated with confirmed choke events, collected over nearly 12 months from February 2019 to January 2020, were assessed to identify distinctive features of growing chokes in the inspection points and sewer maintenance holes. Using these features and retrospective assessment of the data from observed events, we formulated a continuous data scanning methodology to find time periods containing irregularly high water levels, namely the Finding Irregular Duration (FID) method, for early detection of choke events. Application and evaluation of the proposed methodology shows potential in the prevention of wastewater overflows.

The paper is structured as follows. Firstly, the IoT-based smart sewer level monitoring system established by SA Water in Stonyfell, South Australia is described. This is followed by discussion of the observed wastewater level change characteristics in the sewer inspection points and maintenance holes including the behavior before and after a choke event. Next, we introduce the FID method,

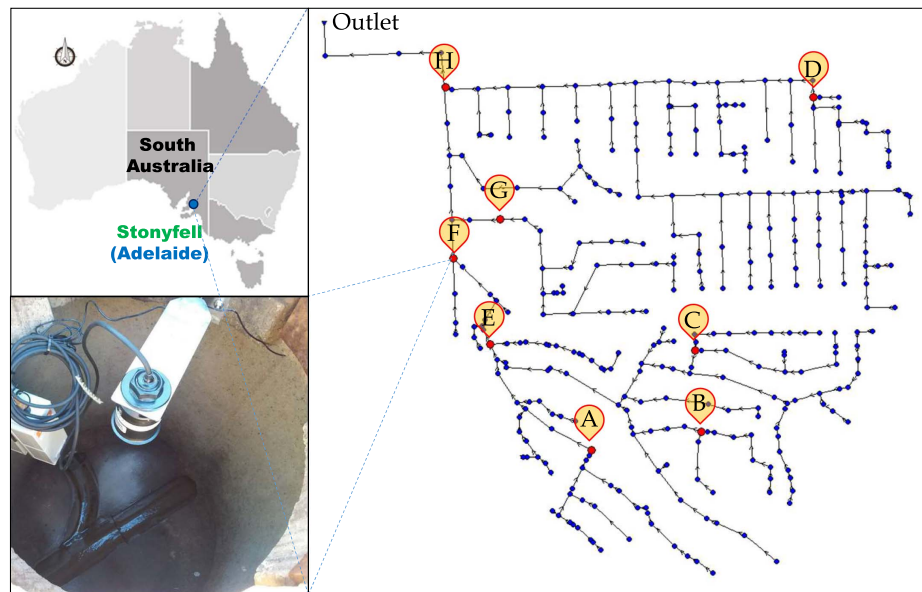


Fig. 2. Experimental sewer network for IoT level sensing system. Eight maintenance holes from A–H were selected for plotting surface water levels and for illustration of proposed methodologies in monitoring choke events.

which is then illustrated through application to selected data sets. Finally, conclusions and suggestions for future work are given.

Experimental Smart Sewer System in South Australia

A pilot smart wastewater system for overflow prevention was installed in the Stonyfell network through 2018–2019 (Fig. 2), which is located in the foothills of the city of Adelaide, South Australia. Due to steep topography, this wastewater pipe network has relatively high gradients, with an average slope of approximately 7.8%. The diameter of the pipes within the system is typically 150 mm, with several 100- or 200/250-mm pipes (the relatively small pipe diameters are because this reticulation network is near the top of a catchment). The total pipe length in the catchment is approximately 11,300 m. The majority of the pipes are vitreous clay (VC) and unplasticized polyvinylchloride (uPVC).

The sewer pipes can be accessed for inspection, cleaning and maintenance through approximately evenly spatially distributed inspection points and maintenance holes which range in depth from 0.7 to 4.1 m (average depth of 1.56 m). Inspection points do not provide particularly convenient access with a typical diameter of 0.15 m. Nevertheless, a number of ultrasonic sensors have been installed at these locations. Sewer maintenance holes provide more convenient access with a typical diameter of 0.6 m allowing for human entry (following confined space entry safety protocols). The upstream pipe and downstream pipe through a maintenance hole are connected by a half-circular open channel, with its diameter equal to the pipe diameter, i.e., typically 0.15 m.

The historic performance of the Stonyfell sewer gravity reticulation network was the primary reason for its selection as the pilot location. The choke rate for this network as of May 1, 2018, was 260 chokes/100 km/year, compared with the average choke rate of 26 chokes/100 km/year over South Australia at the same time. Furthermore, 69% of these historic chokes resulted in overflows to the environment or private properties. The number, types, and placement of sensors for the pilot was determined while taking into account several factors including historic performance, pipe gradient, accessibility, and total project budget.

Fig. 2 shows the sewer pipe network in Stonyfell, South Australia, where 82 ultrasonic water-level sensors were initially deployed. However, only 62 sensors remained operational for the observation period of interest (approximately 1 year of recorded data). Fig. 2(c) shows an ultrasonic level sensor installation in a maintenance hole. The devices used in this installation were Pulsar dBi 6 ultrasonic level sensors (Taggle Systems Pty Ltd, Sydney New South Wales, Australia), coupled with logger and communications module using a 4–20-mA analogue signal. The sensors were installed directly above the center of the flow stream using a laser to achieve alignment.

Each level sensor was installed and configured using vendor's proprietary software. The setup process for these sensors involved training the echo tracing and configuring/calibrating the measurement and analogue signal output parameters. Echo tracing was used to filter out false measurements resulting from sound reflections from known sources (e.g., a bench or a ladder). Analogue signal output parameters were configured to integrate the sensor with the logger and allow for the measurement to be made available and interpretable by users. The two key parameters included an empty distance (i.e., the distance from the face of the transducer to the invert of the pipe/channel) and the span of the 4–20-mA signal (for these devices, the span was set to the maximum distance, which is the empty distance minus a blanking distance). Measurement parameters were set such that the sensor will ping (i.e., sending an impulse signal) a measurement every second, and the output value will be the average of the previous 15 pings (to smooth the measured data).

Following sensor setup, its accuracy was verified using physical measurement. When the sensor was powered up, it functioned continuously as described previously; however, the power supply to the sensor was controlled by the logger. The logger was configured to capture a measurement every 5 min. When the logger needs to capture a measurement, it powers up the ultrasonic sensor for a period of 30 s, and then captures the instantaneous 4–20-mA output signal from the sensor at the end of that period prior to powering down and remaining on standby until subsequent measurement periods.

Given the relatively steep network (and lack of overlap between the range of levels measured by each sensor) and the intermittency

of wastewater discharges from households, as well as the fact that the measurements were not time-synchronized between stations (due to limitations in the logger and communications technologies utilized), each sensor was treated independently to those adjacent (although one or more nearby sensors can provide important operational context to operators assessing data at a particular location and the potential for a downstream choke).

Wastewater level alarms are set in SA Water's Smart Networks Analytics Dashboard (SNAD) system, which is one component of a Microsoft Azure-based information technology (IT) system architecture that SA Water has established for its pilot smart wastewater system (Dix et al. 2021). Two fixed threshold values of 150 and 300 mm were set at each sensor for a low- and high-priority alarm, respectively, for each independent sensor location. Once an alarm is raised in SNAD, a technician manually evaluates the data validity and then raises a request to verify and clear the choke. Since commissioning, this system has been effective in identifying approximately 40% of the wastewater pipe choke events that have occurred in the Stonyfell network (which is considered a relatively high percentage given the sensing system only covers an estimated 48% of the entire Stonyfell network). However, the number of low-priority alarms was very high (>100,000 in 2019) and did not result in any operational action (or even expectation of operational action given the very high number of low priority alarms). Without the low priority alarms, no other methodology exists to proactively identify the early signs of blockages. Therefore, this research focuses on developing methods for early detection of choke events to allow maintenance (cleaning) to be scheduled to proactively prevent future severe choking.

Variation of Water Level in Sewer Maintenance Holes

Significant variation exists in the behavior of the wastewater levels (i.e., the depth of the flow) at different points in the network, complicating the analysis. There are various factors influencing this variation, which include, but which are not limited to, the flows, network topography, upstream and downstream pipe gradients, solid waste or object presence in the maintenance hole during the time of measurement, local hydraulic behavior (i.e., hydraulic jumps), and possible measurement error from the ultrasonic sensors. To demonstrate the challenges in monitoring wastewater levels in a sewer maintenance hole, we have selected eight maintenance holes across the Stonyfell network (labeled following the direction of flow where the notation *A* is for the furthest upstream point and the notation *H* is for the closest to the outlet of the gravity system, as shown in Fig. 2), and plotted their daily level patterns. Data sets associated with these maintenance holes were used afterward in this study to illustrate our proposed monitoring method. In this study, all data sets were preprocessed to remove unreliable data. Specifically, periods where the data show a sudden jump between two consecutive data points larger than 1,500 mm were removed from the data sets (this was based on hydraulic analysis of the network, showing the water level in a manhole cannot increase more than 1,500 mm within the 5-min interval in most scenarios).

Figs. 3(a–h) combine observations from a random week (7 days from a year of data) into a 24-h wastewater level pattern in each maintenance hole. The wastewater level patterns (represented by continuous lines in Fig. 3) were defined by averaging the data within 1-h window across all 7 days. It can be seen that the wastewater level pattern was different at each maintenance hole. The distance of the maintenance hole from individual connections and the randomness associated with the discharge patterns of each individual connection influenced the levels and the degree of variation observed. A diurnal discharge pattern can only be observed for the

downstream maintenance holes, for example, the Maintenance hole H in Fig. 2, toward the outlet of the catchment network (close to the outlet where the effect of the randomness of individual connection flows is aggregated). Fig. 3(h) shows higher flow periods from 7:00 to 9:00 a.m. and 6:00 to 9:00 p.m., with a low-flow period from 2:00 to 5:00 a.m. This pattern was consistent with the typical water demand pattern from domestic users in South Australia urban areas (Do 2017; Marchi et al. 2012; Thyer et al. 2011).

The wastewater levels in the maintenance holes typically range from 20 to 75 mm, showing that wastewater generally flows within the connection channels in the base of the holes. Wastewater levels in Maintenance hole F are an exception to this, regularly reaching levels of around 200 mm (50 mm above the base benchmarking level). This abnormal water level may be due to a possible choke in the downstream pipe that reduces hydraulic capacity and causes the 150-mm outlet pipe from the maintenance hole to be slightly pressurized.

The dynamic characteristics of the network are illustrated by numerous irregular peaks or pulses in the wastewater levels within these maintenance holes. Given the uncertainties associated with a real system, wastewater hydraulic models are unable to predict this dynamic behavior to a high degree of accuracy. Consequently, hydraulic modeling results have had limited practical utility in wastewater overflow monitoring and choke detection applications. Our proposed methods, therefore, are developed based on our domain knowledge, a data analysis approach, and examination of wastewater levels before, after, and during confirmed choke events.

Variation in Early Choke Indicators

Between February 2019 and January 2020, 24 choke events were identified through level monitoring and confirmed as blockages requiring cleaning. Analysis of the 24 events showed that the statistical characteristics (e.g., the box and whisker plots in Figs. 4 and 5) of the wastewater level variations prior to overflow events are not distinctive. Most chokes manifested as a rapid rise in wastewater level within the maintenance hole with little or no prior indication of an abnormality. Only a few chokes demonstrated signs of abnormal levels, representing a partial blockage that was changing (typically growing) with time. The characterizations can be confirmed by comparing the level data prior to the choke (in the lead-up period) with the level data post the choke (when the pipe is known to be in a clean state), as discussed subsequently.

Fig. 4 depicts data from a wastewater overflow event on June 1, 2019, in Maintenance hole H. Fig. 4(a) plots 21 days of time-series level data (including 10 days before the choke event, the choke day, and 10 days after the choke event), which show a clear change in the wastewater level statistics. Figs. 4(b and c) show the characteristics of the wastewater level variations 10 days before and 10 days after the choke, respectively, via the box and whisker plots. This shows that the mean wastewater level, as well as the number of outliers during 10 days before the choke, were significantly higher than the water level under normal conditions after the choke was cleared and the pipe cleaned.

In addition to the box and whisker plots, a two-sample *t*-test was implemented to evaluate the statistical difference between the mean of the wastewater level before and after the choke (postcleaning). The *t*-test statistic value in this case equals 55.77 and its corresponding *p*-value is 0.00, which can be interpreted as the mean and variability of the wastewater level between the two groups being significant different.

By contrast, Fig. 5 plots wastewater levels for an overflow on October 30, 2019, in Maintenance hole G, which illustrates almost no statistical indication days before the choke event. Fig. 5(a) plots

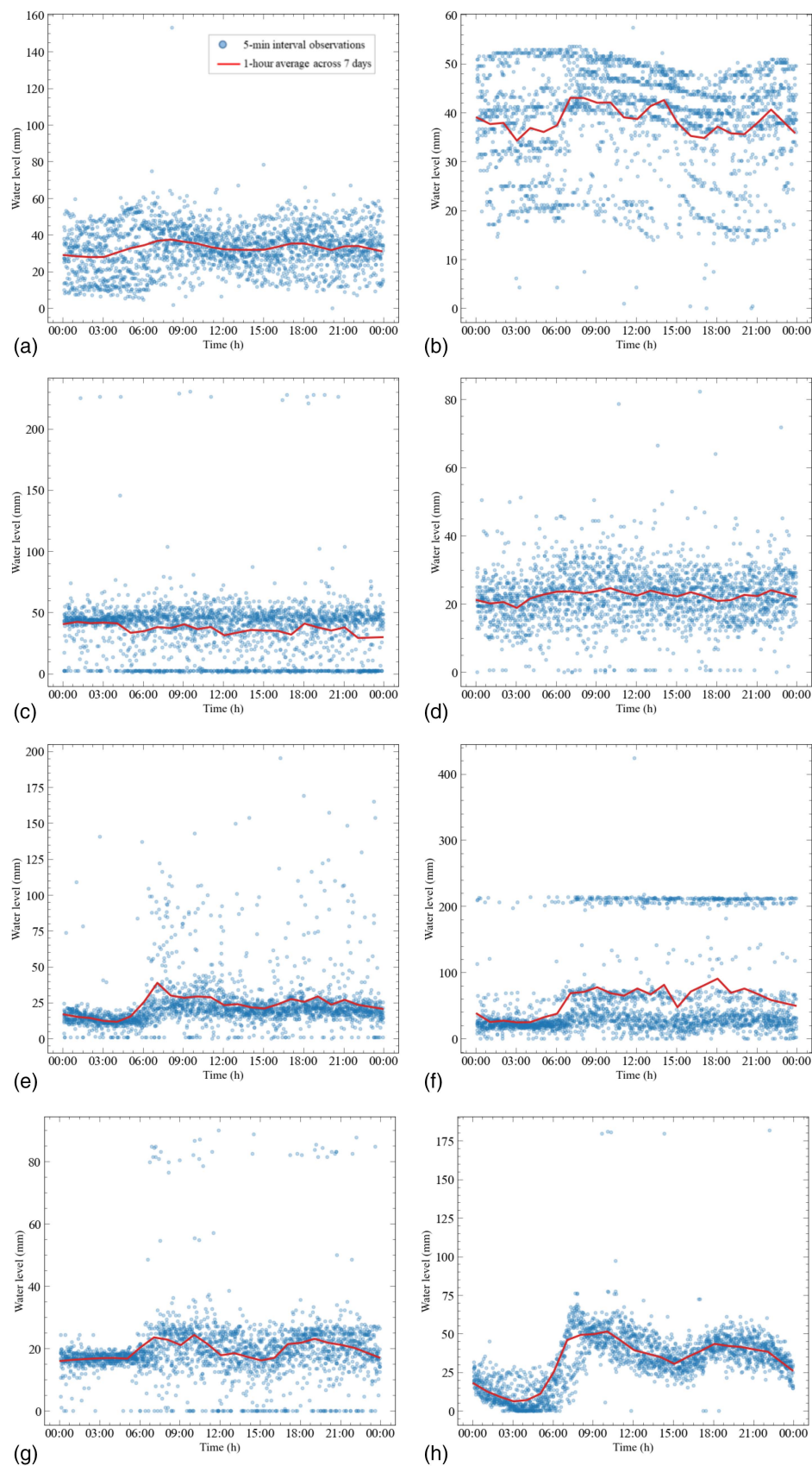


Fig. 3. Data for 7 days overlapped onto a 24-h window to identify water level patterns in the maintenance holes: (a) Maintenance hole A; (b) Maintenance hole B; (c) Maintenance hole C; (d) Maintenance hole D; (e) Maintenance hole E; (f) Maintenance hole F; (g) Maintenance hole G; and (h) Maintenance hole H.

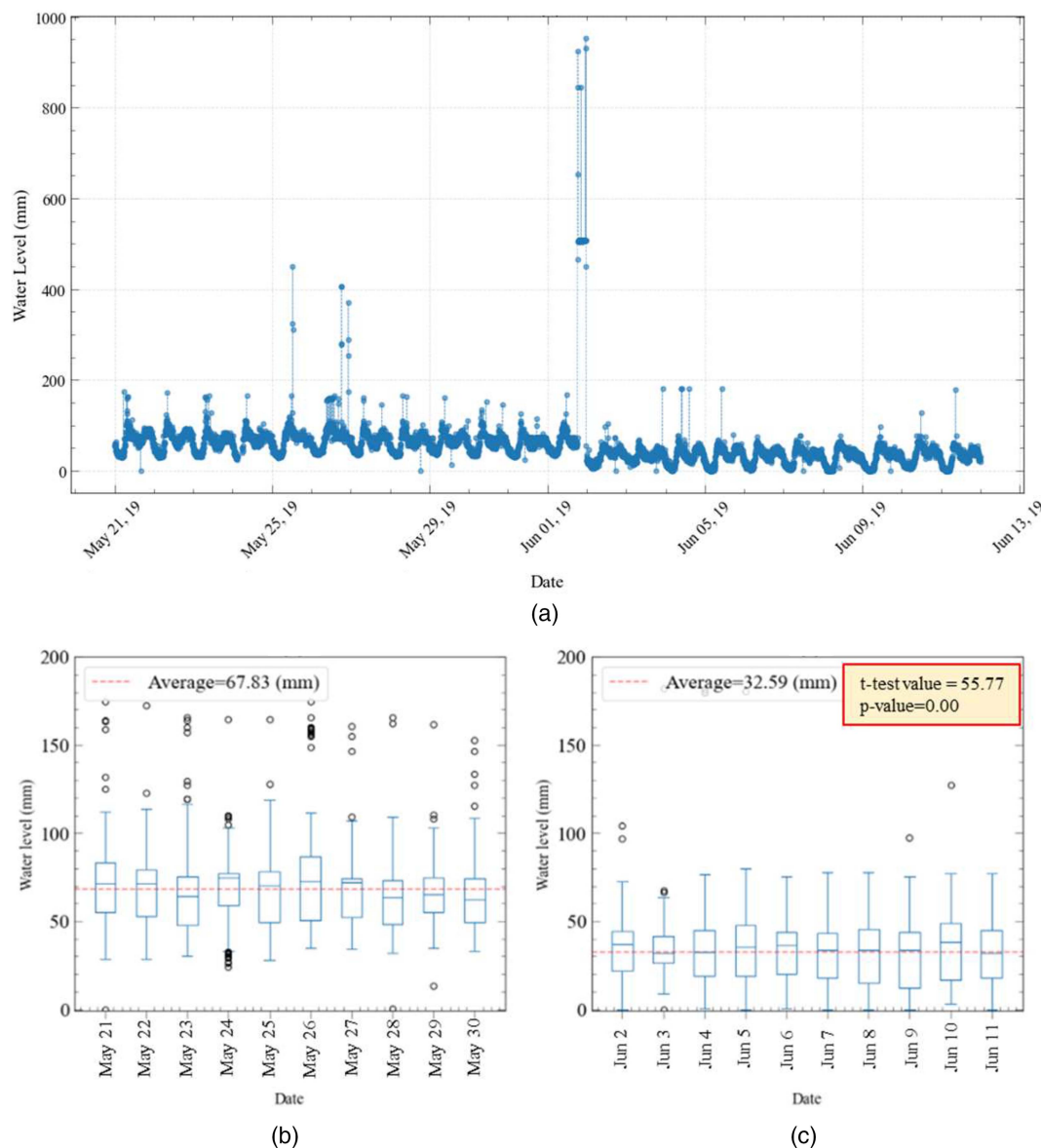


Fig. 4. Wastewater level characteristics from a choke event in Maintenance hole H: (a) sensor data of the event; (b) box and whisker plots for 10 days before the choke event; and (c) box and whisker plot for 10 days after the choke event.

21 days of time-series data, and Figs. 5(b and c) show the box and whisker plots 10 days before and 10 days after the choke, respectively. The two-sample t -test for these two data sets showed no significant difference between the mean wastewater level before and after clearing the choke (the t -test and p -values are 1.34 and 0.178, respectively). This result also highlights the need to have a continuous monitoring system to detect such episodic events with no prior indication.

The clear signs of choking prior to an overflow as shown in Fig. 4 can be detected by standard statistical approaches. However, analyses across the majority of the Stonyfell network data sets have shown that statistical applications to individual or group (i.e., a window) of data points may result in a relatively high number of alarms. This is because the choke forming and the self-clearing process within the sewer network often cause irregular water level patterns (especially at different locations of the catchment). To reduce the number of alarms, statistical approaches would require different sets of parameters for different sensors. Having a single detection method that both considers these temporal and spatial

hydraulic variations to reduce the alarm rates and promptly provide early warning across the network is, therefore, important in terms of data investigation, allocation of resources, and scheduling pipe cleaning work. Although it may not be possible to prevent all sudden chokes leading to overflows, it may be possible to detect a proportion of chokes that gradually build up over time and enable proactive clearing of the blockage. The following section proposes a methodology for the early detection of a choke event.

Methodology for Early Detection of Choke Events

Distinctive Features of Choke

Broadly speaking, blockages reduce the hydraulic capacity of the pipe by reducing the flow rate through the obstruction (e.g., with residual flow either through gaps in the obstruction or between the obstruction and the pipe wall). When the inflow exceeds the hydraulic capacity of the obstructed pipe, potentially most often

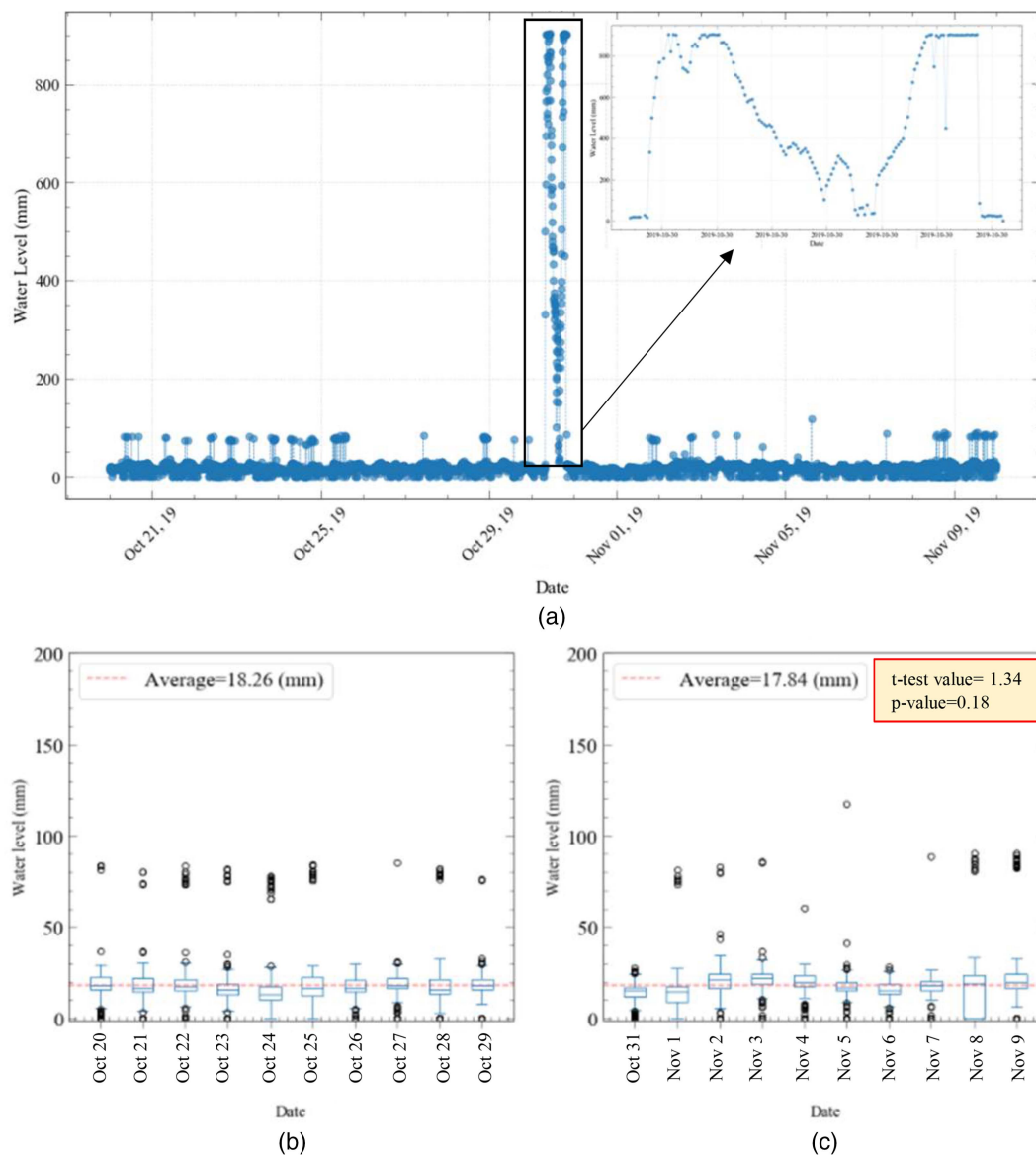


Fig. 5. Water level characteristics from a choke event in Maintenance hole G: (a) sensor data of the event; (b) box and whisker plots for 10 days before the choke event; and (c) box and whisker plot for 10 days after the choke event.

during peak discharge periods, the flow chokes, and wastewater begins to surcharge in the upstream pipe with an accompanying rise in level. Variable storage volume in the network upstream of an obstruction allows the surcharge level at the obstruction to increase and remain high for a period without causing an overflow (with the length of the period proportional to the available storage volume upstream of the obstruction), which is referred to as the holding stage of a choke. During the holding stage, flows within the network transition from free-surface gravity flow to pressurized flow as part (or all) of the upstream pipes become pressurized. When the inflow is smaller than the hydraulic capacity of the obstructed pipe (e.g., during off-peak discharge periods), wastewater drains through and past the obstruction and returns to a more normal level until the next peak discharge period. In some cases, the pressure that builds upon the obstruction may be high enough to either partially or fully dislodge it, allowing wastewater to be quickly released and a return to a state of normal hydraulic operation; these occurrences are referred to as the release stage of a choke.

These processes can repeat until the obstruction develops enough to nearly block the flow leading to an imminent overflow. Therefore, we expect the earliest sign of the formation of an obstruction to be identifiable as slow drainage of wastewater (from only low surcharge levels) due to compromised hydraulic capacity. If the reduction of hydraulic capacity becomes severe enough to affect the hydraulic conditions at the nearest upstream maintenance hole, it will become identifiable as different holding stage followed by release stages of the wastewater.

Within the maintenance hole upstream of the obstruction, the holding stage and release stage will alternate, resulting in anomalies that have two distinguishing characteristics: higher wastewater levels (referred to as the first feature) and wastewater levels persisting at higher levels for longer periods (referred to as the second feature) when compared with historical data (without any obstruction). Here, it is assumed that these historical data represent the normal hydraulic state in the maintenance hole (i.e., historical measurements are used as the benchmark to identify any anomalies against), regardless of the conditions present when measuring the historic

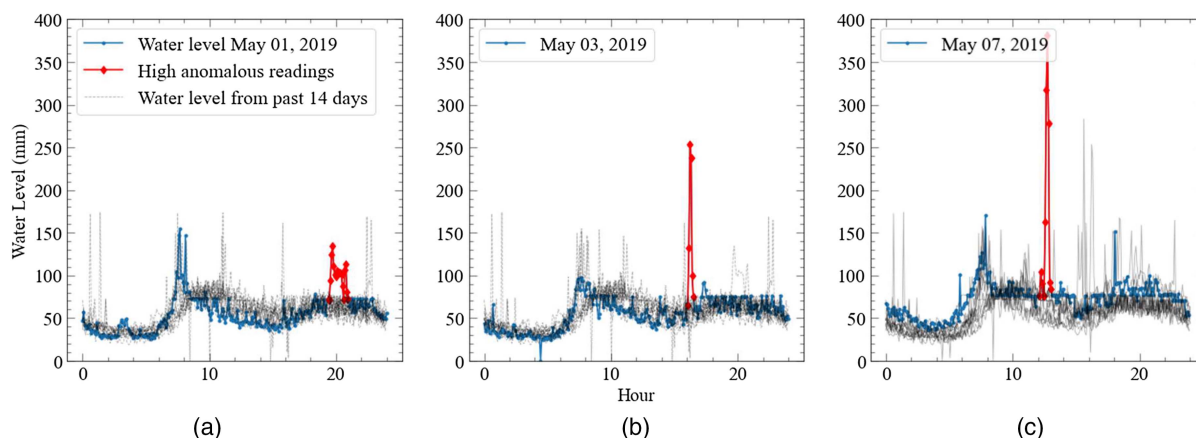


Fig. 6. Examples of wastewater level measurements from Maintenance hole H that have irregularly high level measurements compared with the historic data (i.e., the first indication of a choke forming in the sewer pipe): (a) wastewater level on May 1, 2019; (b) wastewater level on May 3, 2019; and (c) wastewater level on May 7, 2019.

data. Fig. 6 provides three examples of the first feature, which has occurred as the consequence of a progressively growing choke.

In Fig. 6(a), an irregular high wastewater level was observed from 7:25 to 9:05 p.m. on May 1, 2019, approximately 30 days before an overflow in Maintenance hole H, reaching a peak level of 134.35 mm, around 80 mm higher than its peak level over the previous 14 days at the same time of day. Figs. 6(b and c) show two anomalies, one at 4:00 p.m. on May 3, 2019, and one at 12:25 a.m. on May 7, 2019, respectively. The wastewater level for these two events reached 283 and 381 mm, respectively, significantly higher than historical data. These anomalies persisted for successive measurements before returning to normal, which indicates that the obstruction progressed through a cycle of growing and shrinking (self-cleaning), or that at least its geometry changed, during this 7-day interval.

The second feature is related to the duration of irregularly high wastewater levels in the sewer maintenance holes. It is observed that the more severe the blockage, the longer duration that wastewater remains at a higher level than its historical ones. Fig. 7 illustrates these increasing durations that lead to the irregular wastewater level event on May 7, 2019. Fig. 7(a) shows the time series wastewater level data on May 6 and 7, 2019, and Fig. 7(b) shows the durations for which the level remains higher than the reference levels based on historic data from the previous 14 days around the same time of day (i.e., ± 1 h).

The first high period started on May 6, 2019, at approximately 7:55 a.m., and remained higher than the reference levels for a duration of 10 min. The second and third high periods were observed at approximately 5:55 and 6:20 p.m. on the same day, with durations of 15 and 20 min, respectively. This was followed by the final high period, which lasted for 30 min, reaching its peak at around 12:25 a.m. on May 7, 2019. These observable durations, which became longer over time, are clearly an important characteristic for detecting the formation of a choke.

Proposed Early Choke Detection Algorithm

Although the current fixed high-level threshold exceedance approach (generally set at 300 mm) is relatively effective in identifying the first feature, the counterpart fixed low-level threshold exceedance alarm (generally set at 150 mm) results in a high number of alarms, which, due to their high number, currently elicit no response from system operators. It is not known whether responses

to these low-level threshold exceedance alarms would result in earlier detection of a growing choke. This paper proposes a simple method for earlier detection of a growing choke, which is based on the second feature (i.e., wastewater levels persisting at higher levels for longer periods) and may be more relevant and lead to less false alarms than the existing low-level threshold exceedance approach. This method is statistical and targets the second feature described previously, with the aim of accounting for all potential prechoke indicators.

The proposed method is named FID and consists of two steps:

Step 1: Extracting durations over which wastewater levels are higher than normal.

Step 2: Applying a set of rules to raise low priority alarms when the condition for the first step is met.

For Step 1, irregularly high wastewater levels are identified by comparing the upcoming measurements with the 95% exceedance level (L95) value of a subset of reference values (i.e., the level for which 95% of wastewater levels are below). The subset of reference values contains all wastewater level measurements that fall within a 2-h window over the previous 14 days. The 2-h window is identified by projecting to the past 1 h before and 1 h after the relevant data point. The L95 reference was selected instead of the maximum wastewater level to eliminate the intermittent effects of pulse discharges and possible measurement errors, and the 2-h window accounts for temporal variation over the diurnal cycle due to real changes in inflow (e.g., between weekdays and weekends and/or over seasonal change from summer to winter).

Fig. 8 illustrates the process of comparing data for extraction at 6:05 a.m. on a random day. Data from 5:05 to 7:05 a.m. over the last 14 days were collected to identify the L95 reference level. If the wastewater level at 6:05 a.m. is higher than the L95 reference level, then a period of high wastewater level recordings is started. These high wastewater level recordings only finish when the wastewater level drops below the L95 reference level. In this study, the minimum recording duration is defined by at least two continuous data points that are higher than the L95 reference levels (i.e., a minimum duration longer than 10 min).

For Step 2, applying a set of rules to raise low priority alarms when the condition for the first step is met, the following rules are applied:

- The irregular high wastewater level duration needs to be greater than 20 min (i.e., four continuous points).

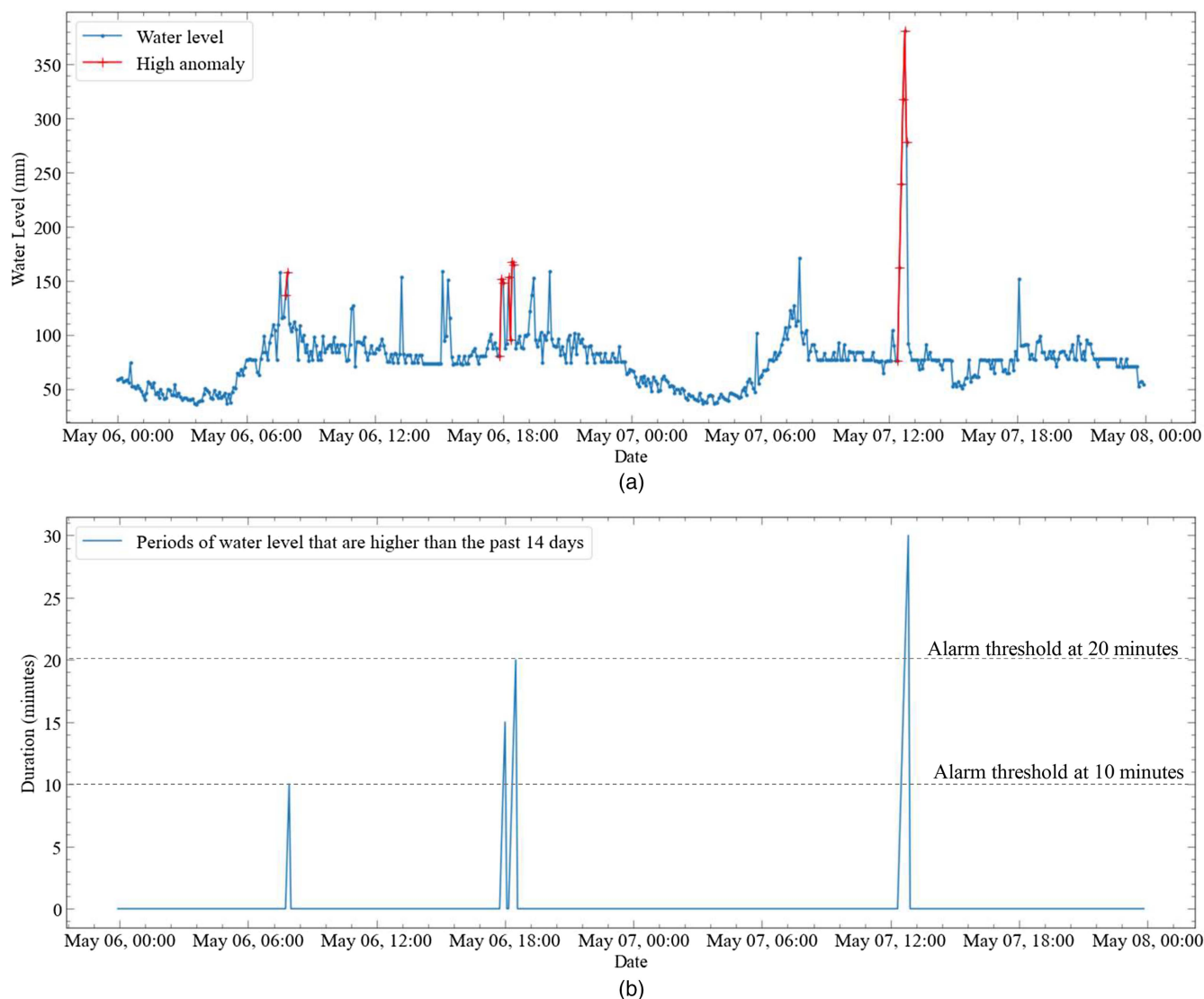


Fig. 7. Examples of wastewater level measurements that irregularly remained at high levels over a relatively long period compared with the historic data: (a) time-series data from May 6 to May 8, 2019, collected from Maintenance hole H; and (b) increasing duration of irregularly high wastewater levels. The dashed lines illustrate the effect of setting the irregular duration threshold to the number of alarms. For example, if the duration parameter is set at 10 min, there will be four alarms, whereas if this parameter is set at 20 min, there will only be two alarms.

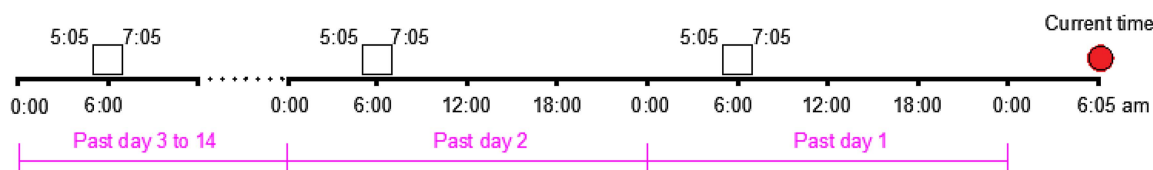


Fig. 8. Data windowing used in the FID method: a 2-h window was selected to compare with data over the last 14 days at the same window time.

- At least one of the irregular wastewater levels needs to be above 1/3 of the diameter of the pipe (i.e., 50 mm for a 150-mm-diameter pipe).

The parameters for the FID method are unique for each sewer system, which should be selected based on understanding of the system and analyses of the system's data sets (and confirmed choke events) in order to balance detection capability and false alarm rates (noting that previous fixed threshold low wastewater level alerts were not responded to). For this Stonyfell sewer network, the

number of alarms generated was evaluated based on combinations of thresholds: (1) at 10, 15, 20, and 30 min, and (2) at wastewater levels of 0 (i.e., without a threshold), 20, 50, 75, and 100 mm. Fig. 7(b) demonstrates how the number of alarms is affected by changes to Parameter 1 when there are four and zero alarms if the duration threshold is set to 10 min and longer than 30 min, respectively, whereas only two alarms will be generated if the duration is set at 20 min. The additional Parameter 2 is used to reduce the number of unnecessary alarms. Any detection derived from this algorithm

triggers a follow-up verification process, through which the presence of a choke is confirmed prior to the generation of any maintenance order.

Statistical Benchmark Methods for Comparison

The performance of the proposed FID method was compared with three other change point detection methods: mean and standard deviation analysis (MSTD), median and interquartile range analysis (IQR), and cumulative sum (CUSUM) change detection. For the first two methods, it was assumed that the maximum wastewater levels within the 2-h window follow a normal distribution. An abrupt change will be detected if the maximum wastewater level in the current window lies outside ($X_t^{\max} \geq \hat{X}_{\text{thres}}$) for the overall pattern of the distribution for the proposed FID method, the threshold $\hat{X}_{\text{thres}}$ is defined based on the same time window over the last 14 days, and in addition

- For MSTD, $\hat{X}_{\text{thres}} = \bar{X} + 3\sigma$, where $\bar{X}$ and σ are the mean and standard deviation of the maximum values for each reference window, respectively.
- For IQR, $\hat{X}_{\text{thres}} = Q_3 + 1.5\text{IQR}$, where Q_3 and IQR are the 75th percentile and interquartile range calculated for all values in the reference windows

The CUSUM method, on the other hand, looks at the accumulated change of the incoming maximum wastewater level observations and detects an outlier if the cumulative sum exceeds a threshold, i.e., as follows:

$$S_t^+ = \max[0, S_{t-1}^+ + X_t^{\max} - X_{\text{Ref}} - k] \geq C_{\text{thres}} \quad (1)$$

where S_t^+ and S_{t-1}^+ = positive cumulative sum for the current window t and previous window $t - 1$, respectively (noting that $S_0^+ = 0$); X_{Ref} = reference point at the current window (for this research, X_{Ref} = mean of the maximum values from past reference windows); k = allowable slack that specifies the size of the shift to be detected; and C_{thres} = detection threshold (i.e., the limit for the cumulative sum to raise an alarm). In this paper, $k = 0.5\sigma$ and $C_{\text{thres}} = 4\sigma$ were selected as common values in various CUSUM applications (where σ is the standard deviation of the maximum values from past reference windows).

Evaluation of the Proposed Method

Evaluation of the Proposed Method on Choke/Overflow Events

The FID method, as well as the number of alarms it generates, was evaluated for 24 confirmed events and 62 available data sets. Table 1 presents the results of the FID method over 24 events in terms of generating its first alarm prior to the chokes/overflows. It is seen that the FID method was able to predict all the events in advance. The number of days that the first alarms were generated ahead of the event ranged from 8 to 57 days with an average of 32.6 days, which was enough for data analysis and investigation before scheduling priority cleaning work orders. Fig. 9 plots selected events to show the FID accuracy, adaptability to different type of chokes, and the necessity to track any potential blockages that may occur in the sewer pipes.

Fig. 9(a) shows the Choke event G (corresponding to the Maintenance hole G), which resulted in an overflow occurring at 8:15 a.m. on October 30, 2019. It was shown previously that there is no significant statistical difference between wastewater levels from 10 days prior to and post the choke event. Furthermore, the dashed horizontal line at 150 mm illustrates that no measurements

Table 1. Evaluation of the FID method over 24 choke/overflow events in terms of its first generated alarm and the number of days prior to the events

Event	First alarm date	Event date	No. of days
A	November 16, 2019	December 26, 2019	40
B	May 21, 2019	July 2, 2019	42
C	Mar 30, 2019	April 23, 2019	24
D	March 5, 2019	April 23, 2019	49
E	April 23, 2019	June 12, 2019	50
F	March 27, 2019	April 19, 2019	23
G	October 8, 2019	October 30, 2019	22
H	April 25, 2019	June 1, 2019	42
J	October 6, 2019	December 2, 2019	57
K	April 19, 2019	May 9, 2019	20
L	November 30, 2019	January 11, 2020	42
M	June 16, 2019	July 2, 2019	16
N	November 15, 2019	November 23, 2019	8
O	January 17, 2020	February 10, 2020	24
P	January 22, 2019	February 9, 2019	18
Q	May 1, 2019	June 2, 2019	32
R	November 8, 2019	November 21, 2019	13
S	December 17, 2019	February 1, 2020	46
T	November 8, 2019	December 19, 2019	41
U	December 22, 2019	January 20, 2020	29
V	October 10, 2019	November 22, 2019	43
W	December 1, 2019	January 9, 2020	39
X	August 26, 2019	September 26, 2019	31
Y	March 27, 2019	April 27, 2019	31

exceed the existing low-level fixed threshold prior to the choke. For this challenging scenario, the proposed FID method raised 11 alarms, with the first alarm raised at 9:45 a.m. on October 8, 2019, i.e., 20 days before the overflow event. The 24-h wastewater level patterns from September 23 to October 7, 2019, are plotted in the inset within Fig. 9(a), highlighting the irregular high wastewater period and showing the water level from 9:45 to 10:50 a.m. on October 8, 2019, being consistently higher than the wastewater level of the previous weeks. This validates the relevance of the first warning from the method (i.e., it was not a false alarm).

In Fig. 9(b), the FID method was applied to the Choke event H (Maintenance hole H) to evaluate its ability in detecting the earliest shift in wastewater level. In this case, the confirmed choke event was recorded at 11:20 p.m. on June 1, 2019. However, a series of irregular peaks (i.e., the first feature) had occurred one month prior, since May 1, 2019, as discussed previously. The FID method provided the first warning at 7:35 a.m. on April 29, 2019, as a result of a persistent period of irregularly high wastewater levels, lasting for approximately 40 min each (highlighted in the subfigure). This initial detection was subsequently followed by a series of alarms throughout the period from May 1 to May 7, 2019 (detailed plots of these excursions shown in Fig. 6 confirm the relevance of these detections). The FID method again raised a series of alarms on May 27, 2019, 3 days before the overflow event.

Fig. 9(c) illustrates an unusual circumstance where the wastewater level before the choke event was persistently high. This suggests that the pipe was possibly partially blocked before the installation of the ultrasonic sensor. The choke event was confirmed on July 2, 2019, in Maintenance hole B, although there likely was another event on June 3, 2019, during which the wastewater level reached 661.1 mm at 1:10 p.m. and high-priority alarms were raised by the 300-mm threshold alarm setting. In this case, the FID method provided the first warning on May 21, 2019, followed by a series of warnings before the unconfirmed choke on June 3, 2019. The last warning was raised on June 20, 2019, around 12 days before the confirmed choke event.

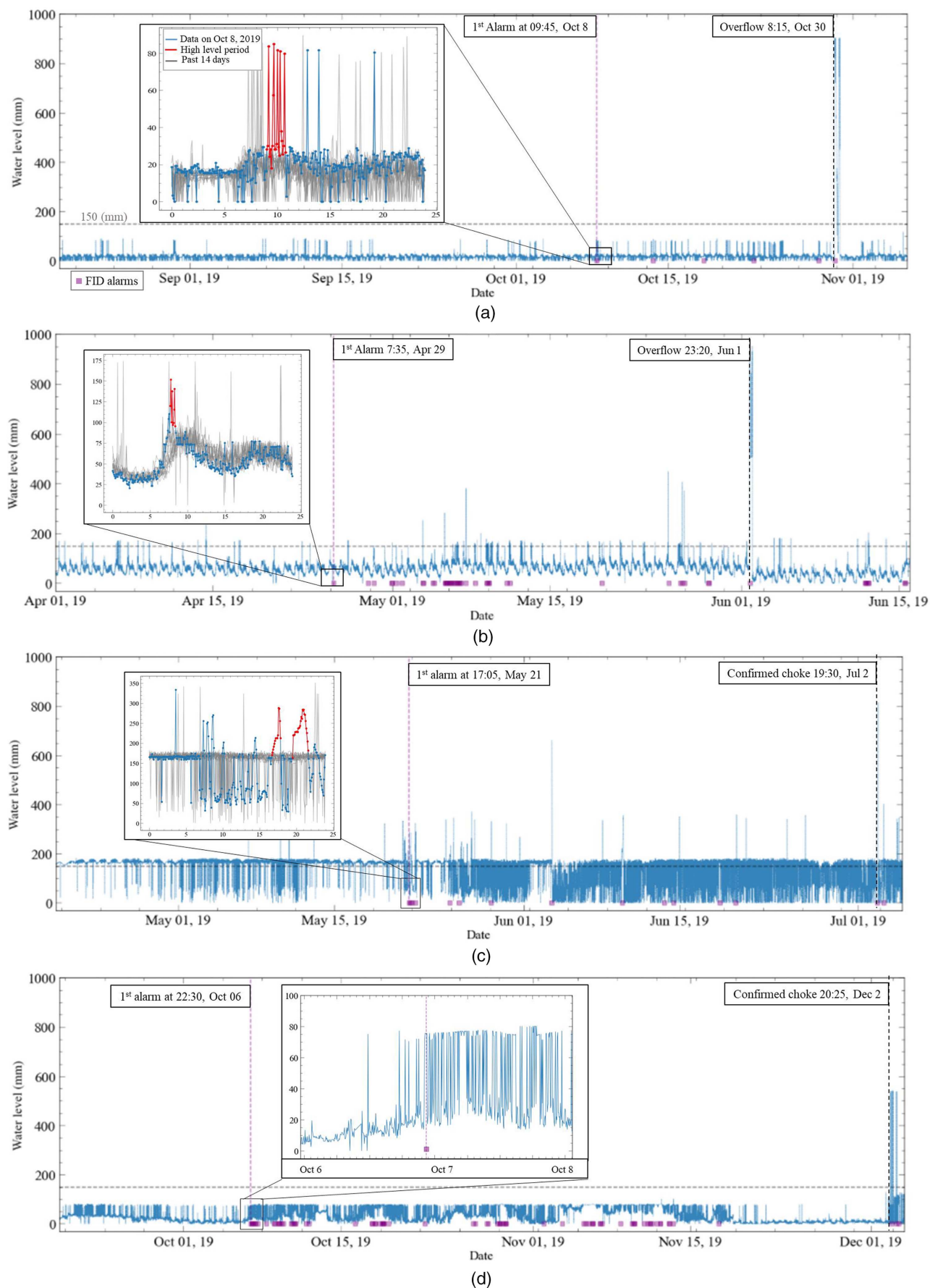


Fig. 9. Performance of FID method over four choke events, namely G, H, B, and J, respectively, with FID detections shown at base of plots: (a) Choke event G; (b) Choke event H; (c) Choke event B; and (d) Choke event J.

Table 2. Comparison among FID method, MSTD, IQR, and CUSUM change detection over five maintenance holes' data sets

Maintenance hole	Comparison period	Average number of alarms per month			
		FID	MSTD	IQR	CUSUM
A	April 1–December 1, 2019	14.4	15.4	35.4	12.5
C	May 1, 2019–January 1, 2020	6.3	8.8	23.6	7.0
D	January 1, 2019–January 20, 2020	2.2	15.8	40.3	10.2
E	January 1, 2019–January 20, 2020	8.3	11.1	31.9	11.0
F	June 1, 2019–January 20, 2020	8.4	18.3	38.1	11.7

Fig. 9(d) plots the Choke event J, which had the longest number of days from the first FID alarm until the event was confirmed. The first alarm was raised on October 6, 2019, 57 days before the confirmed choke event on December 2, 2019. It is seen from the figure that a partial blockage was formed at this time and caused a clear shift in the water level. The partial choke and irregular water levels remained for a significant time from October 6 to November 21 (46 days) before returning back to a normal level. A sudden choke happened 11 days after without any indication within this time. It was surmised that the blockage had been pushed along the pipe and reformed at another location downstream of the original one. Wastewater surcharge due to this new partial blockage did not affect the water level at its upstream maintenance hole, which is represented by a normal water pattern for 11 days. This blockage was only able to be detected again when the pipe became fully blocked after 11 days, when the sensor recorded a significant rise in the water levels at approximately 600 mm. This complicated process highlights the need for detecting and tracking any individual blockages that may form in the sewer pipes for investigation and scheduling cleaning work orders.

Evaluations of the Proposed Method versus Benchmark Methods

The FID performance has also been compared with that from alternative standard statistical change point detection strategies. Five data sets were selected to illustrate the performance of the FID method versus the statistical benchmark methods in this paper, including the MSTD, IQR, and CUSUM methods. Data sets from Maintenance holes A, C, E, and F included confirmed choke/overflow events, whereas there was no overflow event included for Maintenance hole D. In an operational context, good alarm performance is judged by the ability to generate a limited number of alarms that are relevant to valid anomalies. Table 2 summarizes the number of alarms generated by the four methods.

In this case, it is seen that all methods were able to detect confirmed choke events. More importantly, the FID method generated far fewer alarms than the MSTD and IQR methods over all data sets and fewer alarms than the CUSUM method for most of the cases except the data set for Maintenance hole A (14.4 alarms/month for the FID method compared with 12.5 alarms/month for the CUSUM method). The performance of the FID method was shown to be superior, especially given that it had a much greater potential to trigger an alarm because it was determined for every single data point, whereas the comparison methods all operated on subsets of 2-h window data. The FID method was triggered by the persistency of irregularities over 20 min (i.e., 1/3 h) and so had six times as many opportunities to trigger alarms than the other methods.

Fig. 10 illustrates the comparison between the FID method and CUSUM method for Maintenance holes A and D. Fig. 10(a) shows

that both the FID method and CUSUM method were able to detect obvious changes to wastewater level patterns related to the potential presence of a partial blockages (e.g., between March 30 and April 9, 2019). Fig. 10(b) plots a slowly changing wastewater level for Maintenance hole D over the period from February to May 2019, indicating a possible slow growing choke. During this period, both the FID method and CUSUM method were able to detect the changes associated with this trend. However, the FID method generated only five alarms compared with 40 detections from the CUSUM method. Given no verified/confirmed overflow event occurred in Maintenance hole D over an entire year, the number of alarms that FID method generated (2.2 alarms/month) was clearly more reasonable and more reliable than the number of alarms from the CUSUM method (10.2 alarms/month). Such relatively small number of alarms (generated by the FID method) allows operational staff to evaluate the choke state so that relevant responses (e.g., continuing to observe or require immediate cleaning) can be scheduled as required.

Evaluation of the Proposed Method on Its Predictive Accuracy

The accuracy of the proposed model was evaluated using 24 events and all data sets over 3 months from June 1 to August 31, 2019. The current implementation of the FID method targets low-priority alarms (which have been problematic in number based on a low threshold trigger). Its parameters were set to identify any potential blockages that may form in the sewer pipes for further investigation. From analyses of the 24 events, it is seen that a forming blockage may take an average of 32.6 days to cause a severe choke event, in which a series of alarms can be raised within this average blockage development period. Therefore, the predictive accuracy of the proposed method should be evaluated on a set of alarms rather than a single one. In this paper, a false positive alarm was defined if one of the following rules was triggered without having any clear evidence of a blockage within 33 days from the triggered rule: (1) more than two alarms per day, (2) more than two alarms in 2 consecutive days, or (3) more than six alarms per month. On the other hand, an alarm was defined as a false negative if the aforementioned rules were triggered fewer than 3 days before the choke event.

Fig. 11 shows the confusion matrices resulting from the FID method [Fig. 11(a)] and the CUSUM method [Fig. 11(b)]. There were 41 choke events and 54 nonchoke events for this evaluation. Apart from 24 confirmed choke as well as 24 nonchoke events (i.e., the data sets after the confirmed blockages were cleared), the remaining suspected blockages and nonchoke events were selected based on statistical analysis (i.e., the *t*-test and box and whisker plots) from entire available data sets. It was not practical to directly inspect all suspected individual blockages due to resource limitations and the complexity of the CCTV inspection process.

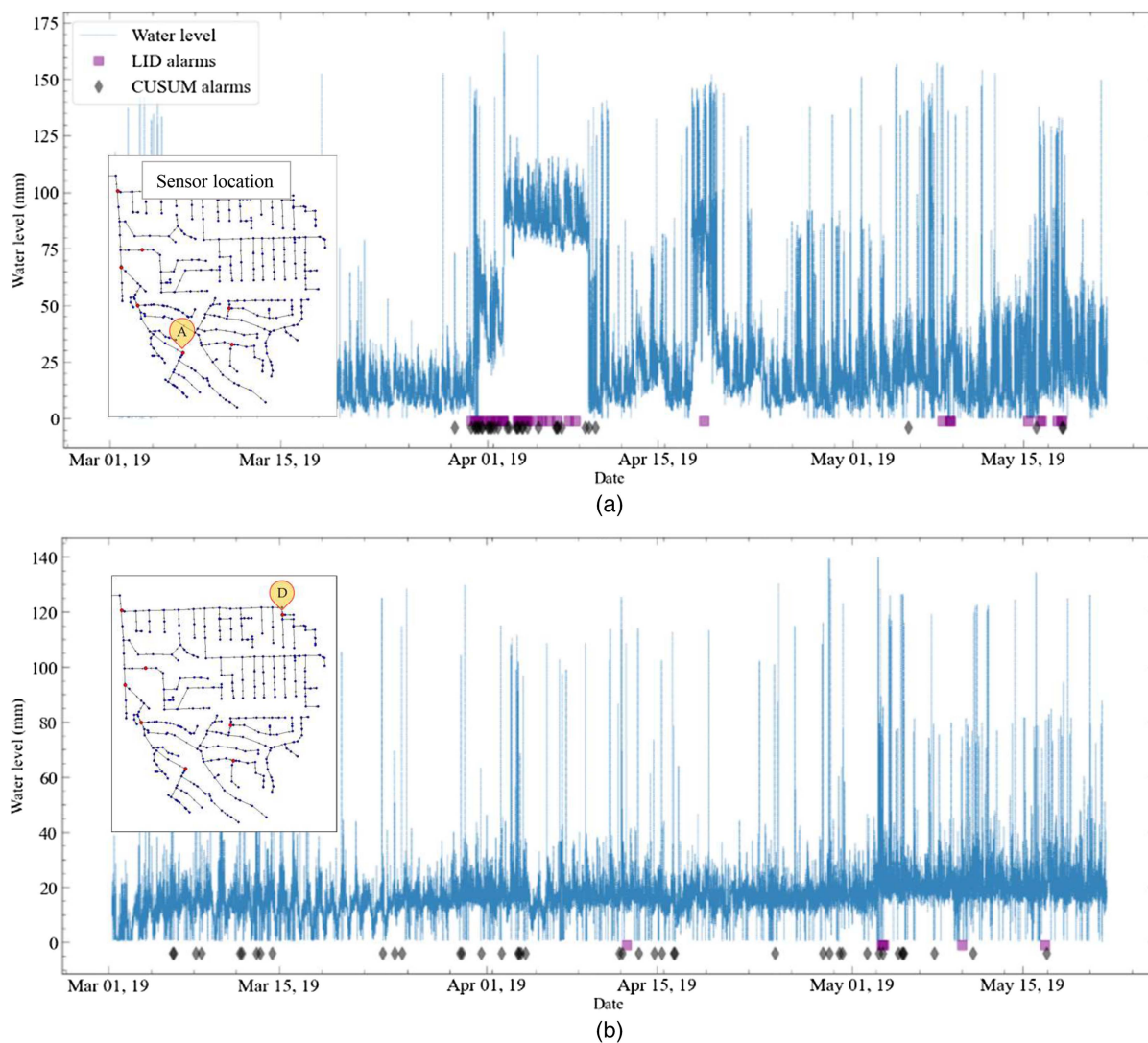


Fig. 10. Comparison between FID method and CUSUM method over Maintenance holes A and D: (a) ability of both the FID method and CUSUM method to detect irregular wastewater levels; and (b) better performance of the FID method over CUSUM method in terms of generating fewer false alarms.

Class	Actual True	Actual False	Class	Actual True	Actual False
Predicted True	40	17	Predicted True	41	42
Predicted False	1	37	Predicted False	0	12

(a) (b)

Fig. 11. Confusion matrices of the (a) FID; and (b) CUSUM methods.

It is seen that the FID method was able to predict 40 out of 41 choke events in advance, whereas the CUSUM method was able to predict all choke events, corresponding with 97.6% and 100% true positive rates. However, out of 54 nonchoke events, the FID

generated 17 (i.e., 31.5%) false positive alarms compared with 42 (i.e., 77.8%) false positive alarms from the CUSUM method. The true negative rate for the FID method, in this case, was 68.5%, and significantly higher than 22.2% for the CUSUM method.

Therefore, the FID method is a superior tool for the early detection of wastewater overflow events.

Conclusion and Recommendations

The installation of ultrasonic level sensors in wastewater networks creates opportunities for monitoring the hydraulic behavior associated with wastewater blockages. The application of statistical tools to data from these sensors can detect the formation of blockages, providing the opportunity to operationally respond and potentially avoid wastewater overflows from the sewer system.

This research examined the characteristics of wastewater levels in an operational sewer network under various operational conditions. Distinctive features in the wastewater level patterns resulting from the presence of a blockage were identified for 24 data sets relating to confirmed choke events, including irregular peaks/spikes in the wastewater levels and persistent periods of irregularly high wastewater level (above historic measured levels). Based on this last observation of persistent periods of irregularly high wastewater levels, an automatic method for the early detection of choke events was formulated, namely the FID method (i.e., finding irregular high wastewater level durations). The FID method was tested for all confirmed choke events, as well as for other available data sets, with results from three events and five data sets reported in detail in this paper to illustrate the performance of the FID method.

In addition, the FID method was compared with three statistical methods: MSTD, IQR, and CUSUM change detection to evaluate its contextual performance. The FID method was able to accurately identify small changes in wastewater level patterns, which can be linked with the early formation of chokes, and outperforms alternative methods in terms of detection accuracy and lower false alarm rates.

Implementation of the FID method in conjunction with the fixed threshold alarm approach (i.e., alarm is raised if the wastewater level rises above 300 mm), provides alarm coverage over the broad range of hydraulic patterns that have been observed, and balances the timeliness of alarm generation with the rate of false alarms. Based on the observed performance, alarms generated by the FID method should be treated as being a precursor indicator of the need for an investigation to assess the conditions within the pipe and/or maintenance hole within a reasonable timeframe (results indicate typical lead times range from days to weeks ahead of major choking events). This highlights the usefulness of using continuous level monitoring (in this study, ultrasonic level sensors) for monitoring, detecting, and preventing wastewater overflows in sewer systems.

Given the description of the targeted features, future work involves investigation of combining multiple techniques as well as machine learning algorithms for a more robust detection approach once the amount of labelled data and confirmed choke events are sufficient. In addition, the potential of the proposed method in generating higher priority alerts and its application to sewer systems with different network topologies should also be evaluated.

Data Availability Statement

Some data and code that support the findings of this study are available from the corresponding author upon reasonable request. These include the codes of the proposed methodology and selected subsets of data from the choke events listed in Table 1, subjected to the

Water Utility's approval. Some data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions. These include the water level data sets.

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